The Impact of Credit Rating Agencies on Financial Markets

LORENZO MENNA (BANCO DE MEXICO) AND MARTIN TOBAL (BANCO DE MEXICO)*

Abstract

The question of whether credit rating agencies (CRAs) influence financial markets has been widely debated in the academic literature, policy circles and general press. However, answering the question empirically generates great challenges because CRAs consider market developments in their decisions, making them endogenous, and financial markets anticipate CRAs' announcements. In this paper, we follow a recent trend in macroeconometrics and argue that credible identification of the impact of CRAs on the markets can be achieved using the high frequency instrumental variable-local projection (IV-LP) methodology. Applying this approach to Mexico, we find that CRAs have statistically significant effects on CDS spreads, interest rates and the exchange rate.

Keywords: credit rating agencies, financial markets, instrumental variable

JEL Classification: G14, F40.

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1. Introduction

The question of whether credit rating agencies (CRAs) have an impact on financial markets has been widely discussed, not only in the academic literature but also in policy circles and the general press (Ashcraft et al., 2011; BIS, 2008; Demyanyk and Van Hemert, 2011). Answering this question empirically generates great challenges. CRAs consider market developments in their actions, making them endogenous: for example, adverse financial shocks can lead CRAs to downgrade a debtor. In this case, the effect of the downgrade is confounded by the adverse financial shock. Moreover, financial markets can anticipate CRAs' actions, resulting in underestimation of the impact of CRAs. In general, identification of causal effects is difficult.

In this paper, we propose using the high-frequency instrumental variable-local projection approach (IV-LP) to identify exogenous variation (i.e., structural shocks) in CRA actions and assess how they affect financial markets. We apply our analysis to Mexico –a country with deep and liquid financial markets compared to other Emerging Market Economies (EMEs)-. The strategy involves using variations in the Mexican sovereign CDS in a small window around CRAs' rating announcements. This variation is linked to the structural shock: CRAs do not make announcements on a particular day based on that day's market information, making endogeneity to market developments unlikely, and market discounting of CRAs' announcements can be plausibly ruled out because it is already priced in the CDS before they take place. Using this series as an instrument in an IV-LP model, we are able to identify CRAs structural shocks.

In recent years, the IV-LP and instrumental variable-vector autoregression (IV-VAR) approaches have gained popularity and have been used to identify credible causal effects in various areas, including monetary policy (Gertler and Karadi, 2015), fiscal policy (Mertens and Ravn, 2013), and political risk shocks (Balduzzi et al, 2023), among others. The advantage of this empirical strategy is that it allows for identification without imposing incredible structural restrictions. Our paper is particularly similar to Balduzzi et al (2023), who use the change in sovereign interest rates in the Eurozone on days of significant political events as an instrument to identify political risk shocks.[1] Similar to CRAs' announcements, these events cause a repricing of perceived sovereign default probabilities, which then transmit to financial markets. To our knowledge, we are the first to apply this approach to the study of the impact of CRAs on financial markets.

Our results indicate that CRA shocks have a statistically significant impact not only on the sovereign CDS, but also on the interest rate spread of government bonds, and the Mexican peso-USD exchange rate. Additionally, these shocks affect private sector variables, such as private bond interest rates. These effects are significant in quantitative terms: a shock corresponding to a one-notch rating downgrade increases the sovereign CDS and interest rate spread of government bonds by approximately 80 basis points and results in a 5%

[1] Similar to us, in the working paper version of their paper, Balduzzi et al (2023) used the CDS as an instrument instead of sovereign interest rates.

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depreciation of the exchange rate within a few days after the shock. The impact on corporate spreads is even greater, with a maximum of 100 basis points. However, the duration of these effects is less clear as the baseline estimations suggest that they may dissipate within one week, although the results vary across different versions of the empirical model.

Conceptually speaking, there are several reasons why CRA may affect financial markets, particularly in contexts of asymmetric information and moral hazard (Altman and Rijken, 2004; Millon and Thakor, 1985; Opp et al, 2013). In their seminal contribution, Millon and Thakor (1985) showed that CRA can gain from sharing information when insiders are better-informed than outsiders about the value of their firms (for a counter-argument, see Kiff et al, 2012; White, 2010).[2] Other authors emphasize that CRAs may influence financial markets because they provide certification services. Credit ratings are used to categorize bonds as investment or non-investment grade, which influences institutional demand and market liquidity (Kiff et al, 2012; White, 2010), and to calculate risk-weights that affect regulatory requirements (Kiff et al, 2012). Credit ratings may also serve as coordinating mechanisms in environments of multiple equilibria. Boot et al (2005) refer to a self-fulfilling equilibrium where markets expect a firm to take risk; they require a higher interest rate; and this, in turn, induces the firm to take more risk. They show that this equilibrium disappears when there is a sufficiently large number of investors that incorporate credit ratings in their decisions (for another example, see Holden et al, 2018).[3]

As mentioned, empirical difficulty in testing these theories arises from identification of causal effects. In particular, CRAs and financial markets respond to common factors that move perceptions of both about a debtor and can confound the effect of CRAs' actions. As we explain in detail below, a first step to address this issue is to regress market outcomes against ratings using daily-frequency data: this allows making credible timing assumptions, i.e. that CRAs do not make announcements on a particular day based on that day's market information, to identify variation in CRAs' actions that is not caused by other confounding factors.

However, this approach still faces two empirical difficulties. Firstly, it is necessary to control for expectations about the rating announcement, as financial markets typically anticipate CRA decisions. Failure to measure expectations accurately can lead to attenuation bias, resulting in underestimation of the impact. Unfortunately, capturing market expectations about CRA announcements is challenging since no surveys or market measures of them are easily available. Secondly, it is important to note that CRAs' actions are not limited to rating announcements but also include statements released on announcement days and other actions outside of these days, such as publication of research notes, data reports, interviews, and participation in panels. Since these actions are not easily observable and

[2] Kiff et al (2012) and White (2010) questioned the role of CRA as disseminators of private information arguing that they lag behind in signaling default risk.

[3] In a self-fulfilling model of sovereign debt crisis, Holden et al (2018) show that CRAs foster coordination around an equilibrium where investors coordinate in a pro-cyclical manner that increases sovereign default risk.

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measurable, controlling for them can be difficult, leading to potential bias if their effect is correlated with that of the rating.

The high-frequency IVLP approach, allows us to address both issues without having to accurately measure either rating expectations or non-rating CRAs' actions. This strategy involves two steps, the high-frequency and the IVLP steps. In the high-frequency step, we take advantage of the fact that the CDS is heavily influenced by CRAs in a small window (one day) around rating announcements. As we explain below, variation of the CDS is strictly related to the structural CRA shock, containing both the rating and statement surprises, in this window: market expectations, in fact, are already priced in the level of the CDS at the time of the announcement and the CDS responds to both the rating announcement and the statement inside the window.

In the IVLP step, variation in the CDS on announcement days is used as an instrument in an otherwise standard local projection. This considers that such variation only captures the structural shock on announcement days and not on all days. Given that non-rating actions outside of announcement days can also affect the markets, CDS variation on announcement days is only a noisy measure of the structural CRA shock and, therefore, must be used as an instrument (Gertler and Karadi, 2015 and Stock and Watson, 2018, among others). Moreover, the LP approach allows us to compute impulse response functions to the structural CRA shock, allowing us to observe how financial markets respond over time.

Contrary to what is common in microeconometric studies, in IV-LPs the "instrumented" variable –the shock- is not observed (Balduzzi et al., 2023; Stock and Watson, 2018; Mertens and Ravn, 2013). Consequently, the scale of the shock must be normalized, and an indicator variable has to be chosen (Stock and Watson, 2018). In practice, we proceed as follows. In a first stage regression, we regress the daily change in the CDS, which we use as indicator variable, against our instrument (a variable equal to variation in the CDS on announcement days and zero on other dates). In the second stage, we regress the predicted value of the first stage on several macrofinancial variables for various horizons.

As mentioned, our paper contributes to the IV-LP and IV-VAR literature. As already emphasized, our approach is particularly close to Balduzzi et al (2023) who study how political risk shocks affect financial markets in the Eurozone. Känzig (2021) uses high-frequency market data to identify oil supply news shocks. Mertens and Ravn (2013) use an IV-VAR model to identify shocks to corporate tax rates with a narrative series.

Most contributions in this literature, however, study monetary policy (Gertler and Karadi, 2015; Jarociński et al, 2020; Nakamura and Steinsson, 2018). In this sense, it is interesting to highlight the analogies between the empirical problems that the IV-VAR and IV-LP approach solve in monetary policy studies and those that they solve in our analysis of CRAs: similar to CRAs, central banks respond to economic conditions when they set the interest rate, which makes it hard to disentangle causality between the two, and do not only affect the markets through interest rate decisions, but also through other actions, such as quantitative easing and, especially, communication, which are difficult to measure. Similar to us, the solution consists in exploiting high-frequency identification techniques: Gertler and Karadi

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(2015), e.g., obtain variation in FED funds rate futures in a small window around FOMC announcements, use it a as a measure of exogenous variation in monetary policy on these days and then as an instrument in an IV-VAR model.

Our paper also adds to the empirical literature on CRAs, which has mainly relied on the methodology of event studies, as proposed by Weinstein (1977), Pinches and Singleton (1978), Cantor and Packer (1996) and Hull et al (2004). These studies aim to measure how the value of financial variables, usually CDS spreads, bond interest rates or stock market values, responds to rating announcements. Using this approach, early contributions found ambivalent results. For example, Weinstein (1977) finds no statistically significant effect of rating decisions on corporate bond prices. In contrast, Cantor and Packer (1996) find that, although markets and CRAs make similar assessments of default risk,[4] CRAs' rating announcements affect prices of non-investment-grade issuers. More recent studies, such as Binici et al (2020) and Finnerty et al (2013), find that rating decisions of CRAs affect CDS spreads, especially after controlling for prior outlook or watch status. Similarly, Alsakka and Gwilym (2012) find that CRAs sovereign credit ratings affect a country's exchange rate, especially if this country is an EME.

Similar to our high-frequency IVLP approach, these studies use daily data and, therefore, exploit the fact that CRAs do not make an announcement on a specific day based on that day's market information. However, even if they sometimes condition on prior watch or outlook status, they do not explicitly account for market expectations about the rating variation and, therefore, may be subject to attenuation bias. Moreover, they disregard the role that the rating statement and other CRAs' actions may play.[5]

Even taking into account the differences between our approach and event studies, it is interesting to note that our results are compatible with those that that literature has found concerning emerging markets. In these countries, the response of sovereign CDS and interest rates to the surprise component of rating decisions has been shown to be statistically significant (Ismailescu and Kazemi, 2010). Further, sovereign ratings seem to have important consequences on bank flows from developed to emerging market economies (Kim and Wu, 2011) and affect exchange rates disproportionately in emerging markets (Alsakka and Gwilym, 2012).

The rest of the paper is organized as follows. Section 2 presents the methodology used for the estimation of the effect of CRAs' communication on financial variables. Section 3 describes the data used for the estimation. Section 4 presents the results and Section 5 concludes.

2. Methodology

[4] In particular, both CRAs and the markets tend to consider mostly publicly available information to assess risk, which suggests that the part of ratings explained by CRAs' private information is quantitatively less important.
[5] Usually, event studies check if the variation in a market variable (often the CDS), relative to a benchmark, in a two-day window around downgrade or upgrade announcements is statistically significant.

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In this section, we discuss the empirical challenges associated with estimating the impact of CRAs' actions (both rating decisions and other actions) on financial markets and how we address them. We will rely on the following model throughout:

$$(1) \qquad \Delta Y_t = \alpha(R_t - E_{t-1}R_t) + \delta s_t + \beta z_t + \epsilon_t$$

$$(2) \qquad R_t - E_{t-1}R_t = \alpha_R \Delta Y_t + \beta_R z_t + \epsilon_{R,t}$$

$$(3) \qquad E_{t-1}R_t = \sum_{s=1}^{T-1} \gamma_s \, w_{t-s} + R_{t-T}$$

where ΔY_t is a Nx1 vector of changes in financial variables of interest between $t-1$ and t, R_t the rating, s_t a Sx1 vector of latent variables that capture all non-rating CRAs' actions in period t, z_t and w_{t-s} are Mx1 and Wx1 vectors of other observable and non-observable variables, contemporaneous and lagged respectively. α, α_R, β, β_R, δ and γ_s are Nx1, 1xN, NxM, 1xM, NxS and 1xW vectors of coefficients, respectively, T is the number of periods since the last rating decision, ϵ and ϵ_{R} are an Nx1 vector of structural shocks and the structural shock to the rating, respectively, and $E_{t-1}R_t$ represents market expectations at time $t-1$ about the rating at time t.

Model (1)-(3) illustrates how the rating decision and financial markets affect each other and each of them is affected by other variables z. Market expectations about rating decisions depend on current rating, R_{t-T}, and on information accumulated since the last rating decision, captured by $\sum_{s=1}^{T-1} \gamma_s \, w_{t-s}$. A regression of Y_t against the rating, of course, is affected by both reverse causality and omitted variable bias. Moreover, even though we do not explicitly model the determinants of s_t, other CRAs' actions can also depend on financial market developments and lagged information, giving rise to similar biases.

Henceforth, we will demonstrate the following: in subsection a), we will explain how using a daily data set allows us to impose timing restrictions that eliminate reverse causality and omitted variable bias when estimating the effect of rating announcements. However, in the presence of measurement error in rating expectations, this does not eliminate attenuation bias and does not help us to measure the effect of non-rating CRAs' actions. In subsection b), we will discuss how high-frequency identification can complement daily data and allow us to obtain the effect of the structural CRA shock, i.e. the combined effect of rating announcements and statements, on rating announcement days. However, this is still subject to measurement error as it does not take into account the actions of CRAs outside of rating announcement days. For this reason, we will then explain that the shock obtained with the high-frequency approach has to be used as an instrument in a IVLP framework to correct for measurement error.

a) Timing restrictions

A first step to address omitted variable and reverse causality bias is to use daily data, as this allows us to impose reasonable timing restrictions. Specifically, we can assume that CRAs do not make decisions on a particular day based on that day's market information. Therefore, with daily data, model (1)-(3) can be rewritten as:

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(4) $\Delta Y_t = \alpha(R_t - E_{t-1}R_t) + \delta s_t + \beta z_t + \epsilon_t$

(5) $R_t - E_{t-1}R_t = \epsilon_{R,t}$

(6) $E_{t-1}R_t = \sum_{s=1}^{T-1} \gamma_s w_{t-s} + R_{t-T}$

In this context, an OLS regression of ΔY_t against $R_t - E_{t-1}R_t$ allows recovering α.

However, as emphasized by Gürkaynak and Wright (2013), it is important to accurately measure market expectations about the decision, $E_{t-1}R_t$, for this approach to work properly. In fact, if the researcher has an imperfect measure of market expectations about the rating decision, $\widetilde{R_t} \neq E_{t-1}R_t$, measurement error causes coefficients b in an OLS regression $\Delta Y_t = b\left(R_t - \tilde{R}_t\right) + u_t$ to be biased towards zero. This issue is commonly referred to as attenuation bias (Gürkaynak and Wright, 2013).[6] Essentially, if the fact that the markets can anticipate rating announcements is not correctly taken into account, their impact will be underestimated.

Unlike other event studies, such as those involving monetary policy or macroeconomic news, which have access to reliable measures of expectations based on market or survey data (Gürkaynak et al, 2020), this is not the case for ratings. To our knowledge, there are no market or survey expectations for ratings. One solution to this problem is to control for lags of w_t. If all the variables in w_t are included, expectations can be accurately measured. However, this is difficult to do in practice.

Additionally, daily data alone does not allow us to measure the effect of non-rating CRAs' actions in s_t. If the researcher is interested in the total effect of CRAs' actions and not only in the effect of the rating, in fact, even measuring the true rating surprise gives a biased estimate of the effect of CRAs. More specifically, this bias depends on the correlation between the rating surprise and non-rating CRA actions.[7]

b) High-frequency IVLP approach

High-frequency identification - Another strategy, the high-frequency identification approach, allows us to address attenuation bias without having to accurately measure rating expectations and to measure not only the effect of ratings but also the effect of non-rating CRA actions. This strategy involves taking advantage of the fact that rating decisions are

[6] Assuming that measurement error satisfies the classical assumptions, i.e. that the difference between the measured and the true rating expectation is uncorrelated with either the independent or the dependent variables, and, without loss of generality, that N=1, $plim(b) = \frac{\sigma^2_{\epsilon_R}}{\sigma^2_{\epsilon_R} + \sigma^2_{E_{t-1}R_t - \tilde{R}_t}}\alpha$, where $\sigma^2_{\epsilon_R}$ is the variance of the structural rating shock $\epsilon_{R,t}$ and $\sigma^2_{E_{t-1}R_t - \tilde{R}_t}$ is the variance of the difference between true expectations and their imperfect proxy. With a good measure of expectations, this variance is zero and the OLS estimate b converges to the effect of the rating, α. However, with a bad measure of expectations, measurement error causes the OLS estimate to be biased towards zero.

[7] This bias arises from omitted variable. As we show below, the CRA shock can be defined as $\epsilon_{CRA,t} = (R_t - E_{t-1}R_t) + s_t$ and equation (4) can be rewritten as $\Delta\widetilde{Y_{i,t}} = A_i\epsilon_{CRA,t} + \widetilde{\beta_i} z_t + \widetilde{\epsilon_{i,t}}$. To the extent that actions in s_t are correlated with the rating shock, there will be bias. In a regression $\widetilde{Y_{i,t}} = b_i(R_t - E_{t-1}R_t) + u_{i,t}$, one obtains $plim(b) = A_i + A_i\sigma_{s,\epsilon_R}/\sigma^2_{\epsilon_R}$, with bias given by $A\sigma_{s,\epsilon_R}/\sigma^2_{\epsilon_R}$, where σ_{s,ϵ_R} is the correlation between the rating surprise and non-rating CRA actions.

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made on specific days, which means that certain variables may be more heavily influenced by CRAs on those particular days. To understand this concept, we can rewrite model (4)-(6) as follows:

$$(7) \qquad \Delta\widetilde{Y_t} = \tilde{\alpha}(R_t - E_{t-1}R_t) + \tilde{\delta}s_t + \tilde{\beta}z_t + \widetilde{\epsilon_t}$$

$$(8) \qquad \Delta Y_{1,t} = R_t - E_{t-1}R_t + s_t + \beta_1 z_t + \epsilon_{1,t}$$

$$(9) \qquad R_t - E_{t-1}R_t = \epsilon_{R,t}$$

$$(10) \qquad E_{t-1}R_t = \sum_{s=1}^{T-1} \gamma_s \, w_{t-s} + R_{t-T}$$

where we stack element 1 of vector ΔY_t and ˜ indicates the remaining (N-1)x1 vector.[8] Note that we normalize α_1 and δ_1 to one.[9] The idea here is that in a small window (say, one day) around the rating announcement, $\Delta Y_{1,t}$ is closely related to the structural CRA shock $\epsilon_{CRA,t}$, which is defined as the combined effect of the rating and statement innovation on $\Delta Y_{1,t}$, i.e:

$$(11) \qquad \epsilon_{CRA,t} = R_t - E_{t-1}R_t + s_t = \epsilon_{R,t} + s_t$$

In particular, in this window, the variance of other shocks, z_t and $\epsilon_{1,t}$, is assumed to be small compared to the variance of $R_t - E_{t-1}R_t$ and s_t. In this case, equation (9) on rating announcement days can be rewritten as:

$$(12) \qquad \Delta Y_{1,t} = (R_t - E_{t-1}R_t) + s_t = \epsilon_{CRA,t}$$

for $t \in RA$ where RA is the set of days in which a CRA rating announcement takes place. Using $\Delta Y_{1,t}$ as the structural CRA shock *on rating announcement days* and rewriting equation (8) as $\Delta\widetilde{Y_t} = A\epsilon_{CRA,t} + \tilde{\beta}z_t + \widetilde{\epsilon_t}$, we can obtain the effect of CRAs' actions on financial markets on these days with the following regressions:[10]

$$(13) \qquad \Delta\widetilde{Y_t} = a + b\Delta Y_{1,t} + u_t$$

Interestingly, this allows us to consistently obtain the combined effect of the rating announcement and the statement, i.e. A, even though only on announcement days. As mentioned in the introduction, a possible candidate for $\Delta Y_{1,t}$ is the change in the Mexican sovereign CDS on rating announcement days. This variable is likely to be the most affected by rating

[8] All other variables and coefficients' dimensions are adjusted accordingly.

[9] We can do this because we can always scale the rating surprise and other CRA actions accordingly.

[10] More precisely, coefficient b_i estimate is $\frac{\alpha_i \sigma^2_{\epsilon_R} + \delta_i \sigma^2_s}{\sigma^2_{\epsilon_R} + \sigma^2_s}$, which is a weighted average of the relative (to ΔY_1) response of variable i to the rating surprise and to the statement, with weights given by the variances of the rating surprise and statement shocks. If the effect of the statement is nil, then b recovers the relative effect of the rating surprise, α_i. With equation (8) rewritten as $\Delta Y_{i,t} = A_i\epsilon_{CRA,t} + \beta_i z_t + \epsilon_{i,t}$, in any case b_i is a consistent estimate of A_i.

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announcements and statements, since both credit rating agencies and CDS contracts evaluate the likelihood of a borrower's default.[11]

Note that although the high-frequency approach accurately captures structural CRA shocks on rating announcement dates, it is still susceptible to measurement error. As mentioned in the introduction, CRAs also communicate with the market outside of these dates, meaning that s_t is not zero even outside of rating announcement windows. Therefore, the structural shock on announcement days is only a noisy measure of the true structural shock.

High-frequency IVLP model - To address the issue that the high-frequency approach only provides a noisy estimate of the true structural shock, we follow a recent trend in macroeconometrics and combine it with a standard LP (Gertler and Karadi, 2015; Balduzzi et al., 2023; Känzig, 2021). Specifically, we use the noisy measure of the structural shock, in our case the change in the CDS on rating announcement days, as an instrument in an IV-LP. The LP framework also has the advantage of enabling us to study the transmission of the shock over time.

Compared to a typical instrumental variable-microeconometric study, the IV-LP model presents an additional challenge because the structural CRA shock, $\epsilon_{CRA,t}$, cannot be directly observed. As a result, the effect of this shock can only be determined up to a scaling constant (Stock and Watson, 2018). However, this issue can be addressed by normalizing the scale of $\epsilon_{CRA,t}$ and assuming that a unit increase in $\epsilon_{CRA,t}$ corresponds to a unit increase in a policy indicator variable.[12] By doing so, the outcome variables can be expressed in terms of an observable regressor in the following way:[13]

$$(14) \qquad \Delta\tilde{Y}_{t+h} = a_h + b_h \Delta Y_{1,t} + u_{t+h}.$$ [14]

where h indicates the horizon over which the LP is estimated and $\Delta\tilde{Y}_{t+h}$ is a vector of changes in variables Y between $t+h$ and $t-1$.

Similar to Balduzzi et al. (2023), we use the daily change in the sovereign Mexican CDS *on all days* as our policy indicator. [15] We made this decision based on two main factors: i) a suitable policy indicator has to be strongly affected by the structural CRA shock, which

[11] This is also the reason why event studies on rating announcements often estimate the effect of rating announcements on the CDS of the borrower (Binici et al, 2020).

[12] Note that structural shocks are not only unobserved in IVLPs, but in all VARs and LPs. Also in these frameworks, a normalization is routinely applied. This normalization often involves setting the variance of the structural shock to one, rather than its contemporaneous effect on a policy indicator (Stock and Watson, 2018).

[13] To be more precise, this assumption allows expressing the indicator variable as the sum of the structural shock of interest and a linear combination of other contemporaneous and past structural shocks, $\{\widetilde{\epsilon_t}, \epsilon_{CRA,t-1}, \widetilde{\epsilon_{t-1}}, \dots\}$, $Y_{1,t} = \epsilon_{CRA,t} + \{\widetilde{\epsilon_t}, \epsilon_{CRA,t-1}, \widetilde{\epsilon_{t-1}}, \dots\}$.

[14] $u_{i,t+h}$ is a linear combination of contemporaneous structural shocks, $\epsilon_{CRA,t}$ excluded, and the leads and lags of all structural shocks, i.e., $u_{i,t+h} = \{\epsilon_{CRA,t+h}, \widetilde{\epsilon_{t+h}}, \dots, \widetilde{\epsilon_{,t}}, \epsilon_{CRA,t-1}, \widetilde{\epsilon_{t-1}}, \dots\}$.

[15] Balduzzi et al (2020) instrument the daily change in the CDS with the daily change in the CDS on days of events related to political events to identify political risk shocks.

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makes the instrument relevant and ii) it should be measured in "the native units relevant for policy analysis," as stated by Stock and Watson (2018).

Regarding i), we previously mentioned that the CDS is the variable most likely to be affected by CRA actions. Regarding ii), a better choice could be the rating, as it seems more natural to measure the effect on financial markets of a rating change, rather than of an increase in the CDS caused by a CRA shock. Of course, however, the rating does not respond to non-rating CRA actions, which is incompatible with i). Therefore, we opted for the CDS as our policy indicator. Our choice is similar to that of Gertler and Karadi (2015) in the context of monetary policy shocks: they use interest rates on government bonds instead of the natural monetary policy indicator, the Fed funds rate, as the former responds to forward guidance.

Then, we can instrument our policy indicator $\Delta Y_{1,t}$ (the daily change in the CDS) with the instrumental variable Z_t, which is formally defined as:

$$(15) \qquad Z_t = \Delta Y_{1,t} \quad if\ t \in Rating\ announcement\ days; \quad Z_t = 0 \quad otherwise.$$

i.e. a variable equal to the change in the CDS on rating announcement days and equal to zero outside. Requirement for the instrument to be adequate are:

$i)\ E\left(\epsilon_{\mathrm{CRA},t} Z_t\right) = \mathrm{B} \neq 0$ (relevance);

$ii)\ E\left(\widetilde{\epsilon_t} Z_t\right) = 0$; (contemporaneous exogeneity);

$iii)\ E\left(\widetilde{\epsilon_{t+j}} Z_t\right) = E\left(\epsilon_{CRA,t+j} Z_t\right) = 0$; $\forall$i for $j \neq 0$ (lead-lag exogeneity).

These three conditions state that the instrument must be correlated with the contemporaneous structural CRA shock (the relevance condition), but not with other structural shocks, both contemporaneous and lagged (respectively, the contemporaneous and lead-lag exogeneity condition). These requirements are similar to those that make the high-frequency identification valid, i.e. the change in the CDS on rating announcement days must be closely related to the structural CRA shock $\epsilon_{CRA,t}$ (the relevance condition) and that the variance of other shocks, background noise, is small (the contemporaneous and lead-lag exogeneity conditions).[16]

In this context, it is important to address our use of a daily window surrounding rating announcements. The choice of window size involves a trade-off between smaller and larger windows. While larger windows allow more time for the market to assimilate the

[16] The presence of background noise in the instrument makes the estimate inconsistent. Define the true model of variable i as $\Delta Y_{i,t+h} = A_h \epsilon_{CRA,t} + \beta_{i,h} z_t + \epsilon_{i,h,t}$ and assume that the indicator variable behaves as $\Delta Y_{1,t} = \epsilon_{CRA,t} + \beta_1 z_t + \epsilon_{1,t}$. Assume moreover that the instrument is contaminated by background noise, i.e. $Z_t = D(\zeta_1 \epsilon_{CRA,t} + \zeta_2 z_t + \zeta_3 \epsilon_{1,t})$ with $D = 1$ on announcement days and zero otherwise. In this case, the IV estimate is $plim(b_{i,h}) = A_h + \frac{(\beta_{i,h} - A_h \beta_1) D \zeta_2 \sigma_z^2 - A_h D \zeta_3 \sigma_{\epsilon_1}^2}{D \zeta_1 \sigma_{\epsilon_{CRA}}^2 + \beta_1 \zeta_2 \sigma_z^2 + D \zeta_3 \sigma_{\epsilon_1}^2}$. The bias is larger when background shocks have larger direct effects on outcome variables than indirect effects through the policy indicator. Noise uncorrelated with outcome variables tends to cause negative bias. A larger correlation of the instrument with the CRA shock tends to reduce bias.

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information, smaller windows are more likely to fulfill the assumption that background noise is negligible. When it comes to high-frequency identification of monetary policy shocks and data releases, existing literature has typically settled on windows of a half- to one-hour duration (Hoek et al., 2020; Gürkaynak et al., 2005).

Regrettably, we do not possess intraday CDS data, which means that we are required to employ a daily window. Nonetheless, we do not believe that a daily window is inherently inferior to a smaller one in the context of CRAs. Other researchers have used daily windows in their analyses of oil news and political risk shocks, as in the cases of Känzig (2021) and Balduzzi et al. (2023), respectively. Similar to these instances, the dates of rating announcements are not known beforehand, which means that markets may be slower in processing the announcement, interpreting it, and reacting to the news (Brooks et al., 2003). Consequently, a daily window may be more advantageous in these scenarios.[17] However, a disadvantage of the daily window is that the CDS is a volatile variable that reacts to other developments in financial markets, which raises the risk that background noise is present.

We approach this issue using two methods. First, we include controls in our IV estimation of equation (14). In the context of equation (9), this means that we make every effort to control for z_t, thereby reducing background noise and isolating variation in $(R_t - E_{t-1}R_t) + s_t$. Second, we conduct placebo tests in which, instead of utilizing the change in the CDS on rating announcement days as an instrument, we employ dates that have been randomly selected. In section 5, we demonstrate that these placebo tests do not yield significant estimates, which suggests that we are correctly identifying CRAs shocks.

In practice, estimating (14) through IV regression allows us to obtain our parameters of interest:

$$(16) \qquad b_h = \frac{E\left(\widetilde{\Delta Y_{t+h}^{\perp}} Z_t^{\perp}\right)}{E\left(\Delta Y_{1t}^{\perp} Z_t^{\perp}\right)};$$

where $\perp$ indicates that a variable is orthogonal to the controls we include in our estimation. $b_{i,h}$ is a consistent estimate of the relative (to ΔY_1) impulse response function of variable $\widetilde{Y}_i$ to the structural shock $\epsilon_{CRA,t}$.

We develop two empirical specifications to study the effects of CRA shocks on Mexican financial markets. The baseline specification includes only the first difference of the sovereign CDS in the vector of outcome variables $\widetilde{Y}_t$. The extended specification, on the other hand, adds the CDS spread of PEMEX, the EMBI+ and the CEMBI spreads (which measure the difference between the interest rate on Mexican sovereign and corporate dollar bonds over US treasury bonds, respectively), expressed in first differences, as well as the Mexican peso-USD exchange rate and the Mexican stock market index IPC, both expressed in log first differences.

As controls, we include past realizations of $\widetilde{Y}_t$ and Z_t, as well as contemporaneous and past realizations of the CBOE Volatility Index (VIX), the Brent crude oil spot price (both in

[17] CRAs event studies usually consider one to two days windows.

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log-differences) and an average of the sovereign CDS spreads of other emerging market and middle-income economies (in first differences). [18] These global controls help ensure that our instrument is not correlated with global shocks that affect Mexican financial markets. We consider a horizon of two business weeks (ten days). For this reason, we include up to ten lags of the endogenous variables and nine lags of the instrument and global controls.

3. Data Sources

We use two sources to construct our dataset. The five-year maturity CDS spreads on Mexico's sovereign debt, PEMEX and the other emerging markets were obtained from IHS Markit.[19] The Mexican Stock Market Index IPC, the Mexican peso-USD exchange rate, the EMBI+ Mexico Sovereign Spread, the CEMBI Corporate EMBI Mexico Blended Spread, the CBOE implied volatility index (VIX) and Brent crude oil spot price were taken from Bloomberg.

To construct our instrument, we identify dates of CRAs' rating announcements on Mexico's sovereign 5-year maturity dollar debt. We consider all the rating and outlook decisions of the three major CRAs, i.e. Fitch Ratings, Moody's, and Standard & Poor's, made from January 4, 2010 to December 31, 2019.[20] Over this period, the CRAs made 38 rating and outlook decisions in total: 15 were from Fitch Ratings, 5 from Moody's and 18 from Standard & Poor's. Of these decisions, 25 were rating confirmations (no change in outlook nor rating), 4 were outlook upgrades, 6 were outlook downgrades, 2 were rating upgrades, and 1 was a rating downgrade.[21] Using the note that CRAs publish on the day of rating announcements, we were able to distinguish the announcements that were made before market closing time from those that were made after.[22]

Then, we construct our instrument as the change in the CDS spread between the end of the day when the CRAs make the announcement and the end of the previous day. In cases where the announcement is made after the market closing time, we use the change in the CDS spread of the following day. Figure 1 displays the instrument, along with the direction of the CRAs' announcement. Bold and light red arrows indicate rating and outlook downgrades,

[18] Emerging and middle-income economies follow the definition of the IMF, see appendix for the complete list.
[19] We use the five-year maturity US dollar CDS contract.
[20] Excluding the global financial crisis and the Covid-19 pandemic from the estimation period makes identification more credible, as periods of high market volatility are not considered. This is particularly relevant, given that three rating and outlook decisions on Mexico were taken in March and April 2020, in the midst of severe market volatility. Our data source for the CDS of the sovereign and of Pemex is Markit, whose dataset covers the period 2010-2021. Notice that no watch decision was taken by any of the three CRAs in the period covered by our estimation.
[21] In total, the dates with rating decisions are 36, since in two days of our sample there were simultaneous decisions. The first date is October 31st, 2018, when Fitch Ratings made a negative outlook change and Standard & Poors confirmed its previous rating. The second is June 5th, 2019, when Fitch ratings downgraded its rating and Moody's made a negative change of outlook.
[22] More precisely, given that CDS contracts are traded in OTC markets, we use as closing time the hour in which Markit collects and publish CDS data from several markets. We were not able to identify the time at which the announcement was made in 14 cases and assigned the announcement to the day of publication. This is unlikely to affect the results: in fact, of the announcements of which we were able to identify publication time, only one was after closing time.

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respectively, while bold and light green arrows indicate rating and outlook upgrades. Events without arrows denote rating confirmations.

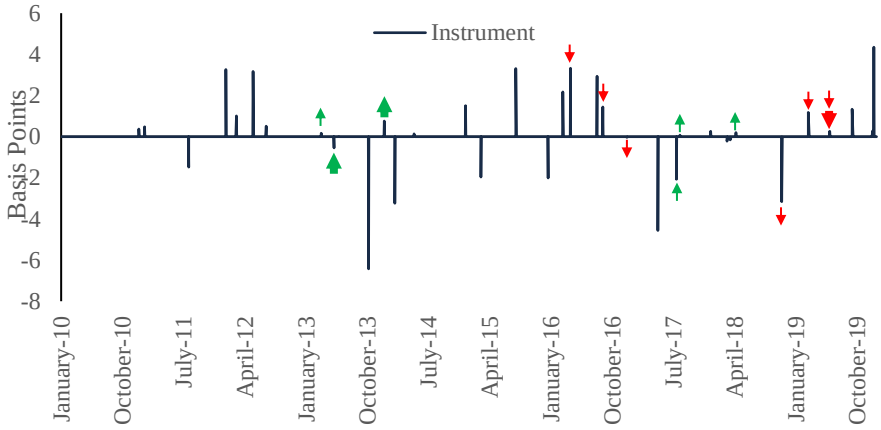

Figure 1 - Source: Elaborated with data from IHS Markit, Fitch Ratings, Moody's, and Standard & Poor's. The instrument, plotted as slim blue bars, is the daily change in the closing sovereign CDS spread of Mexico with respect to the previous day, on the days on which CRAs take decisions about the sovereign rating. The dates were obtained from online publications on the CRAs websites.

Analysis of the notes published by CRAs on announcement days indicates that their rating decisions are based on public information related to both macroeconomic and financial developments. For example, Moody's cited Mexico's "*spending cap on primary current expenditures*" as a reason for upgrading its rating in 2014, while Fitch Ratings highlighted the resilience of the Mexican economy to various shocks, including "*weaker economic growth [and] lower-than-expected oil production*", when it confirmed Mexico's rating on March 16, 2018. This suggests that CRAs update their ratings based on publicly available information, and hence that the markets may be able to anticipate their announcements. Furthermore, even when ratings are confirmed, CRAs provide their assessment of the debtor's financial situation, which can still impact the markets.

In this context, it is interesting to notice that the change in the CDS on the days of announcements does not always go in the same direction as the rating decision, suggesting that indeed the markets may be sometimes surprised positively (negatively) by negative or neutral (positive or neutral) CRAs' announcements.

4. Results

In this section, we report the results of our IV-LP approach, as we mentioned, considering a baseline empirical specification and an extended one. In the baseline specification, we only include the sovereign CDS as endogenous variable and the exogenous controls, i.e. the VIX, Brent price and CDS of other EMEs. In the extended specification, we also include among endogenous variables the CDS of Pemex, the EMBI+ and the CEMBI spreads, the exchange rate and the stock market index.

a) Baseline model

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As a first step, we consider the model with the sovereign CDS as endogenous variable and without global controls.[23] Even though our dataset only contains 36 dates with CRAs' announcements over a sample of 2513 days, the coefficient on our instrument in the first stage regression is positive and highly significant. The robust t-statistics is 15, implying an F-statistics of 237, much larger than the critical value of 10 usually considered in the literature (see, e.g., Gertler and Karadi 2015; Stock et al, 2002). This suggests that our instrument is sufficiently correlated with our indicator variable and that the relevance requirement is satisfied.[24]

Figure 2 shows the impulse response of the sovereign CDS to a CRA shock. As explained in the previous section, the IV-LP methodology requires a scale normalization of the impact response of the sovereign CDS, which we set to 1 basis point. The effect on the CDS increases to more than twice the shock three days after and remains statistically significant more than one week.[25]

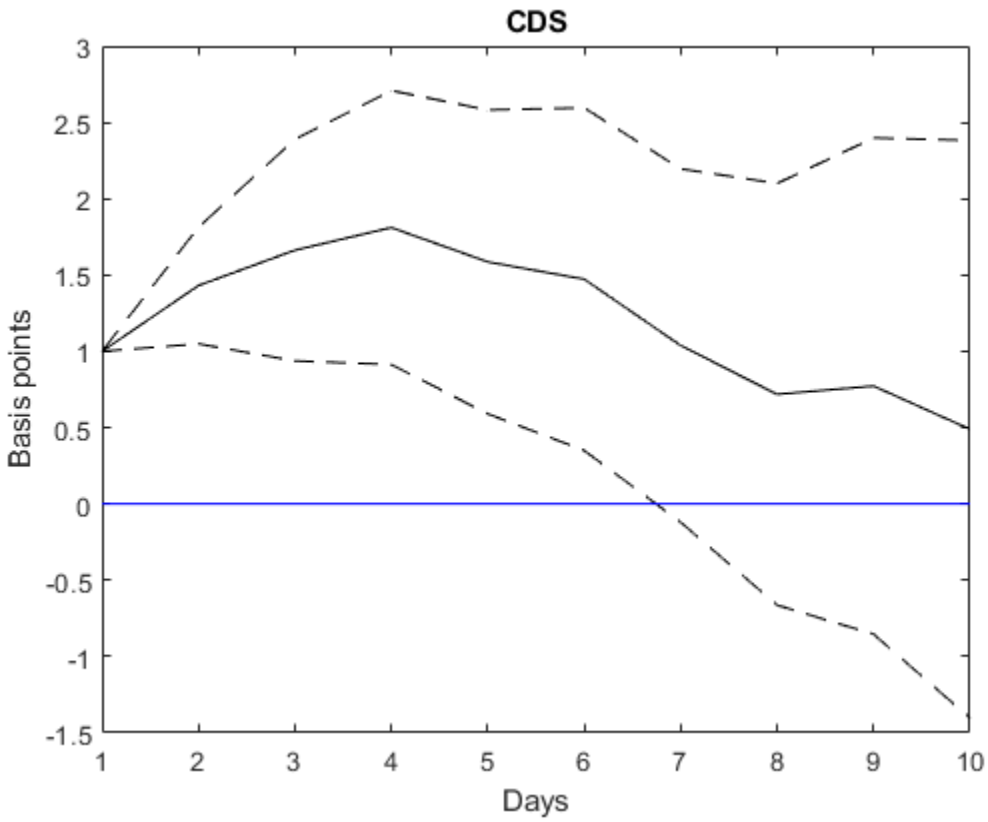

Figure 2 – impulse response of the sovereign CDS in the baseline model, with only the daily change of the CDS as endogenous variable and no exogenous control. Ten lags of the daily change of the CDS and nine lags of the instrument are included. The dashed lines represent the 95% autocorrelation and heteroskedasticity robust Newey-West confidence bands.

[23] As explained in section 2, we also include ten lags of the sovereign CDS and of the instrument as controls.

[24] However, this is also a clue that in this version of the empirical model background noise is substantial. In fact, this strong correlation between the instrument and the policy indicator may be the result of the other shocks causing the CDS to vary. Below, we show that placebo tests do indeed suggest that the baseline model is probably measuring a combination of the effect of CRA and other shocks.

[25] More precisely, the impulse response is significant up to six business days after the shock.

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As discussed in section 2, including controls is a way to reduce background noise. E.g., the CDS is affected by global financial shocks.[26] Likely candidates are shocks to the appetite for risk of global investors, to market perceptions about EMEs and to oil prices. As for other EMEs, in fact, bouts of risk aversion in global markets usually result in higher risk premia on Mexico's sovereign debt. Similarly, Mexican financial markets are exposed to specific emerging market factors that cause co-movement among the sovereign CDS of these countries (see, e.g., Binici et al, 2020). Controlling for oil prices is also important, given that oil revenues represent an average of 25% of total government revenues in our sample and falling oil price may cause the CDS to rise. As mentioned in section 2, we use the VIX, an average of the CDS of EMEs and the Brent crude oil price to control for these shocks.[27]

Figure 3 shows the impulse response functions of the CDS in the model with the VIX (left panel), the Brent oil price (central panel) and the average of the CDS of EMEs (right panel) as controls. In all specifications, the robust F-statistics are well above 10; 41, 70 and 34 respectively. Full lines are the impulse responses and the dashed lines the 95% confidence bands. In all specifications, the impulse response function remains positive and statistically significant for more than one week, with an effect of more than twice the shock in the models with the VIX and the CDS of EMEs as controls.

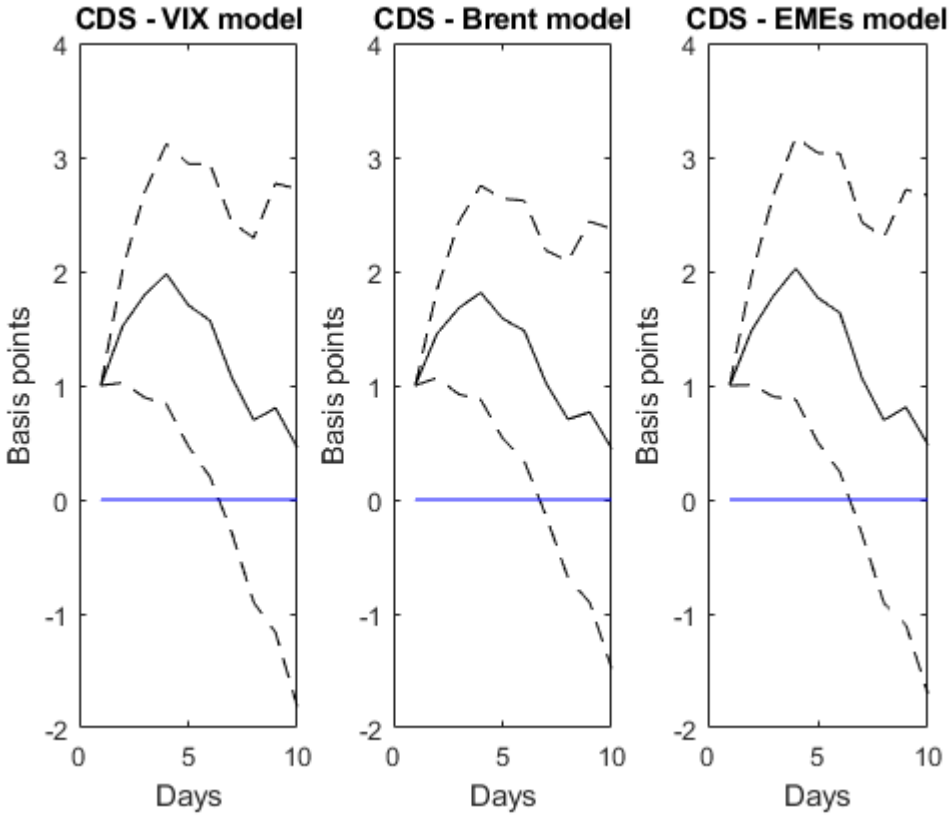

Figure 3 – The left panel reports the impulse response function of the sovereign CDS in the model with the contemporaneous and nine lags of the daily change in the log of the VIX as controls and only the daily change of the CDS as endogenous

[26] Of course, also non-financial shocks, e.g. shocks to output or sovereign debt, may in principle bias the results.
[27] As explained in section 2, we include contemporaneous and nine lags of these variables.

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variable. The central panel reports the impulse response function of the sovereign CDS in the model with the contemporaneous and nine lags of the daily change in the log of Brent price as controls and only the daily change of the CDS as endogenous variable. The right panel reports the impulse response function of the sovereign CDS in the model with the contemporaneous and nine lags of the daily change of an average of the CDS of EMEs as controls and only the daily change of the CDS as endogenous variable. In all models, ten lags of the daily change of the CDS and nine lags of the instrument are included. Full lines are the impulse responses and the dashed lines the 95% autocorrelation and heteroskedasticity robust Newey-West confidence bands.

In the last step, we also estimate a specification including the VIX, the Brent price and the average CDS of EMEs as controls at the same time (F-statistics here is 22). Figure 4 shows the impulse response of the sovereign CDS. As in the model without controls, the effect of the CRAs shock on the sovereign CDS is significant for more than one week. The CDS increases to more than twice the shock on day 3 and remains above the initial shock until day 7.

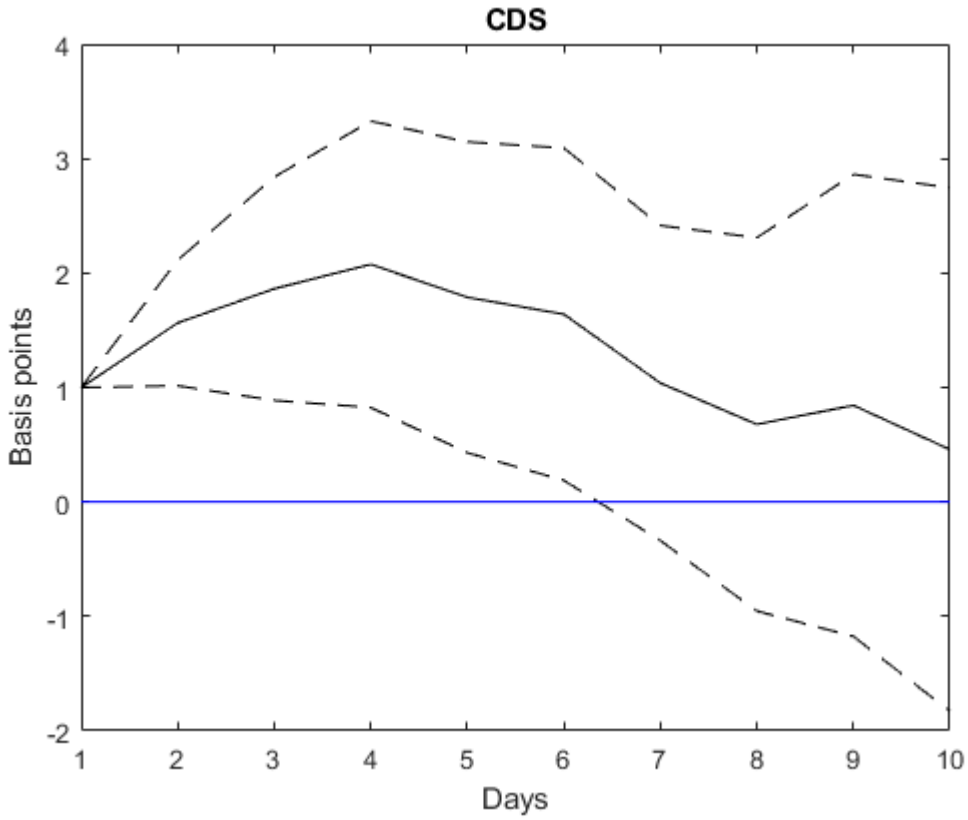

Figure 4 - impulse response of the sovereign CDS in the model with the daily change of the log of the VIX, the daily change of the log of Brent oil price and the daily change of the average CDS of EMEs as controls and only the daily change of the CDS as endogenous variable. Nine lags of the controls and of the instrument and ten lags of the CDS are included. Dashed lines are the 95% autocorrelation and heteroskedasticity robust Newey-West confidence bands.

b) Extended model

CRAs may affect other important financial variables, apart from the sovereign CDS. Though the literature on sovereign ratings has mainly considered this latter variable, the exchange rate and public and private sector borrowing costs have also been analyzed (Alsakka and Gwilym, 2012; Ismailescu and Kazemi, 2010). The IV-LP methodology can easily be used to analyze the effect of CRAs' shocks on these variables in a unified framework.

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Hence, in our extended model, in addition to the global controls, we include the CDS of PEMEX, the EMBI+ and CEMBI spreads, the peso-USD exchange rate and the stock market index.[28] Again, the coefficient on our instrument in the first stage regression is positive and significant (the t-statistics is 4.62 and the F statistics is 21.41, larger than 10). In Figure 5, we present the impulse response functions. Similar to the baseline model, the effect on the CDS is statistically significant for more than one week and increases to more than twice the shock after three days. Interestingly, the shock has an effect of similar magnitude on the CDS of PEMEX: the response on impact is similar and reaches more than twice the size of the shock after three days. The EMBI+ spread response is also close in magnitude and persistence to that of the sovereign CDS. Thus, financial markets definitely seem to consider CRAs' opinions in their assessment of the value of public sector bonds.

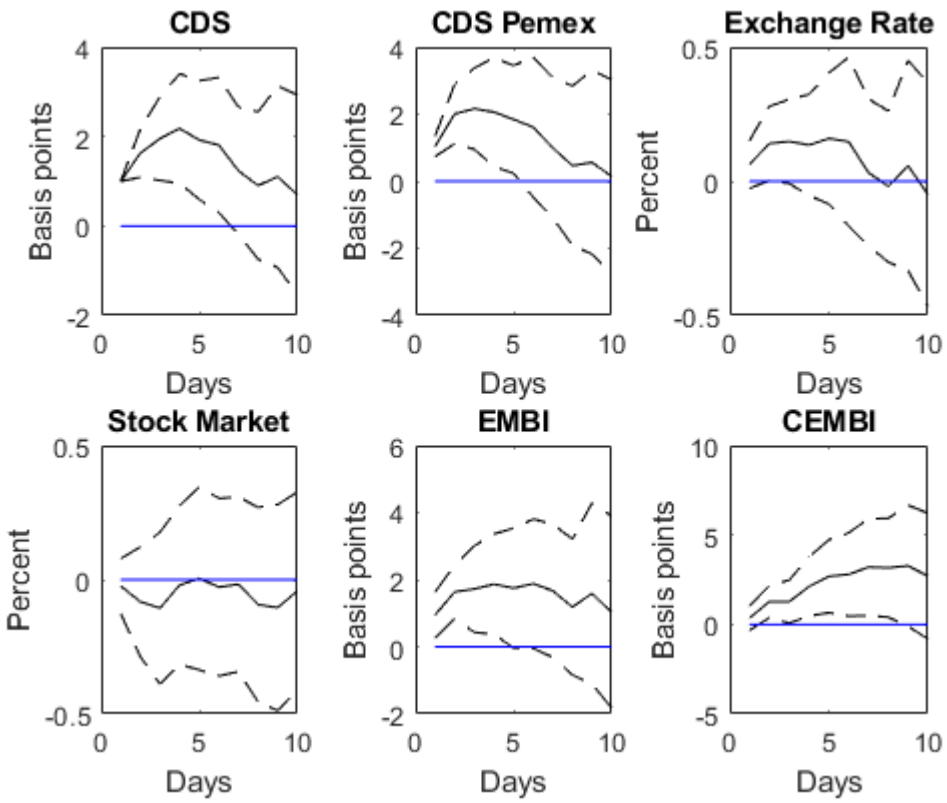

Figure 5 – Impulse responses in the extended model with the daily change of the CDS, the CDS of PEMEX, the EMBI+ and CEMBI spreads and the daily change of the log of peso-USD exchange rate and of PCI Mexican stock markets index as endogenous variables. The daily change of the average of the CDS of EMEs and the daily change of the log of the VIX and of the Brent oil price are included as controls. Ten lags of the endogenous variables and nine lags of the controls and of the instrument are included. The continuous black line represents the impulse response function after a CRA shock that generates an increase in the Sovereign CDS of 1 basis point. Dashed lines represent the 95% autocorrelation and heteroskedasticity robust Newey-West confidence bands.

[28] Again, we include ten lags of these endogenous variables and the contemporaneous value and nine lags of the exogenous controls.

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Interestingly, the shock also causes a depreciation of the peso-USD exchange rate. The response is quantitatively strong (around 0.15% at peak for a shock of 1 bp to the CDS), though statistically significant only the day after the shock. The significance of this response suggests that CRAs can not only have important effects on the borrowing cost of the public sector, but also have wider macroeconomic consequences.

It is also interesting to understand if there is any effect on variables relevant for private sector financing, such as interest rates on corporate bonds and the stock market. As Corsetti et al (2013) shows, in fact, worsening perceptions about the sovereign are not only usually reflected into higher sovereign interest rates, but also affect private sector borrowing costs. Given the systemic relevance of the sovereign, in fact, government debt is often used as benchmark for pricing private sector assets (Wooldridge, 2001). As figure 5 shows, the CEMBI spread is significantly affected by the CRA shock, while the stock market does not. The response of the CEMBI spread is somewhat larger and more persistent than that of the EMBI+ spread, though it becomes statistically significant one day later.

It is important to note that the effect of the shock tends to become not statistically significant within two weeks for all variables. This may be a result of a power problem (Nakamura and Steinsson, 2018). High-frequency identification like the one we use here tends to isolate clean but relatively small shocks. Over time, financial markets are affected by a myriad of other shocks, making the signal to noise ratio too small to be able to make reliable inference. In section 5, we show how efficiency can be improved estimating an IV-VAR instead of an IV-LP. As we show there, in that case, effects are statistically significant for longer.

5. Discussion and sensitivity analysis

In this section, we first discuss how background noise may bias our estimates and the role that controls play in this context (subsection a). Then, we discuss the quantitative aspects of our results (subsection b) and compare them to those that are obtained using a timing restriction approach (subsection c). Then, as further robustness checks, we use an IV-VAR instead of an IV-LP to understand the implications of stronger assumptions in terms of invertibility and shock transmission (subsection d) and estimate the IV-LP model again, including only dates that contain rating or outlook changes in the instrument (subsection e).

a) Accounting for background noise

As explained in section 2, our identification strategy is based on the assumption that on rating announcement days no other shocks apart from the CRA shock affect the CDS, conditional on our set of controls. The CDS, however, notoriously move around a lot also outside of announcement days, which suggests that CRA shocks are far from being the only or main determinant of CDS fluctuations. The shocks that cause movement in the CDS outside of announcement days can of course occur also on announcement days, generating noise. Even though our controls may eliminate a large part of it, it is still important to account for it explicitly.

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To do this, we follow Balduzzi et al. (2023) and run placebo tests. In particular, we estimate our empirical model using as instrument the daily change in the sovereign CDS spread on randomly selected dates, instead of announcements dates. We repeat this procedure 2000 times and present the 97.5 and 2.5 percentiles as intervals. If our controls are indeed sufficient to account for all background noise, these intervals should include the zero.

In regard of the baseline model without global controls (Figure A1 in the appendix), this is not the case: in particular, the CDS response intervals do not cover the zero in the first three days after the shock, suggesting that controls are indeed needed to ensure identification.[29] Adding the Brent oil price to the model does not change this result (Figure A2, central panel): the interval does not include the zero in the first three days after the shock, as in the model without controls. However, including the VIX and, especially, the average CDS of EMEs substantially improves the results. When we control for the VIX, in fact, the interval does not include the zero for only two days after the shock, rather than three (Figure A2, left panel). Controlling for the average CDS of EMEs ensures that the placebo interval includes the zero at all horizons (Figure A2, right panel). A similar result is obtained when including all global controls at the same time (Figure A3).

Concerning the extended model, the placebo test shows that the zero is included in the intervals for all variables, apart from the contemporaneous response of the CDS of PEMEX (Figure A4). This is not surprising, as the sovereign CDS and the CDS of Pemex are closely related and the contemporaneous response basically captures a regression of one variable against the other on a subset of dates.

The placebo tests suggest that background noise is an important concern in the baseline model and that including controls is necessary. Including the VIX and, especially, the CDS of other EMEs in the regressions ensures that the placebo tests do not deliver significant results. For this reason, in both the baseline model with controls and the extended model background noise seems to be satisfactorily controlled for.

b) Quantification

An intuitive way to think about the quantitative effect of CRAs on the markets is to ask what is the effect, e.g., of a rating downgrade on financial market variables. However, the fact that CRAs influence the markets not only through rating decisions but also through non-rating actions makes it difficult to respond to this question. This in fact implies that we have to normalize the shock as to have a unit effect on the CDS, therefore precluding the possibility of directly measuring the effect of a rating change.[30]

However, we can still include the rating in the IVLP estimation as an endogenous variable, normalize its contemporaneous response to the CRA shock and rescale the other impulse

[29] Notice that the contemporaneous response is one by construction, so no confidence band is estimated on it.

[30] As mentioned above, it is in principle possible to use the rating as policy indicator and therefore measure the CRA shock in rating units. However, in this case the instrument does not seem to satisfy the relevance requirement. This is likely related to the fact that the rating is less sensitive to non-rating CRA actions.

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response functions accordingly. Doing this allows us to measure how, e.g., a rating downgrade induced by a CRA shock, i.e. an exogenous downgrade, affects the markets.

To do this, first we define $\Delta R_t^{\perp}$ as the rating change, orthogonal to controls, on day t: in particular, we assume that a rating downgrade (upgrade) is equal to 1 (-1) and an outlook downgrade (upgrade) is equal to 0.5 (-0.5).[31] Then, we include $\Delta R_t^{\perp}$ in vector $\Delta \widetilde{Y}_t$ in regression (14). Figures A5 and A6 in the appendix report the rescaled impulse responses to a rating downgrade in the four versions of the baseline model and in the extended model, respectively.[32]

In the four versions of the baseline model (Figure A5), the response of the CDS on impact is around 40 b.p. and rises to levels between 70 and 80 b.p. after 3 days. This suggests that a rating downgrade leads the markets to price in an increase in sovereign default probability that reaches a peak of about 0.7-0.8%. This increase is largely reabsorbed within two weeks. As explained above, however, the latter could be the result of a power problem rather than a truly temporary effect.

In the extended model (Figure A6), the effect of a downgrade on the CDS is estimated to be similar to that of the baseline model. Similar increases can be observed also in the case of the CDS of Pemex and the EMBI spread. The exchange rate depreciates by around 5% by day 2 (the only day where it is statistically significant), though also in this case the effect seems to be rapidly reabsorbed. As already mentioned, the response on the CEMBI spread is more sluggish, reaching an increase of more than 100 b.p. within day 10.

c) Comparison with a timing restriction approach

As explained in section 2, the advantages of the high-frequency IV-LP approach compared to an approach that only relies on timing restrictions on daily data are that it allows for i) consistently estimating the effect of CRA shocks even without a good measure of rating expectations and ii) obtaining the effect not only of the rating, but also of other CRA actions. To assess the importance of these issues, in this section, we compare the results of the high-frequency IV-LP model to those of an approach that only uses timing restrictions.

To do this, we estimate the following LP:

$$\Delta Y_{t+h}^{\perp} = a_h + b_h \Delta R_t^{\perp} + u_{t+h} \tag{17}$$

Similar to the high-frequency IV-LP model, we estimate both the baseline and the extended models and include ten lags of the endogenous variables and nine lags of the rating change.

As explained in section 2, measurement error in rating expectations causes attenuation bias, i.e. an underestimation of the true effect of a CRA shock. Additional bias, which can go in either direction, arises from the fact that other CRA actions are excluded from the estimation.

[31] When the announcements of two CRAs happen on the same day, we take an average. E.g., if one CRA downgrade the outlook and the other downgrade the rating, the value of the rating variable is 0.75.

[32] We do not report the confidence bands in these figures, as statistical significance is unaffected by the rescaling of the impulse responses.

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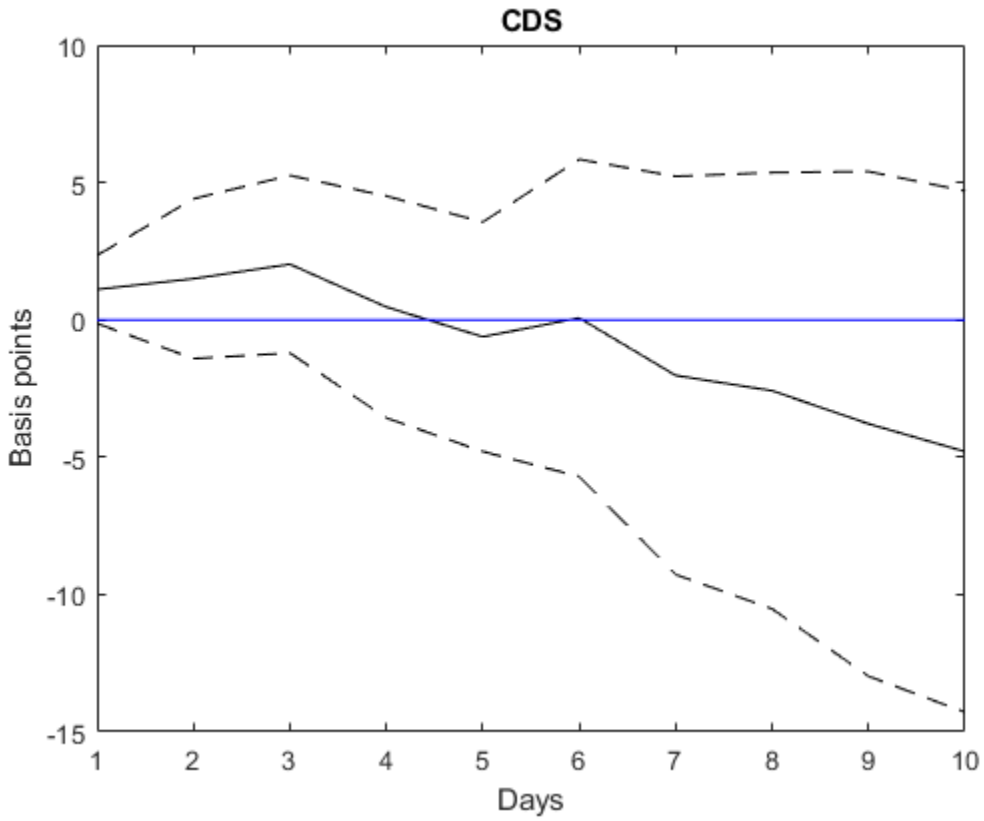
Figure 6 and 7 report the impulse response functions to a rating downgrade in the baseline and extended LPs, respectively.[33] In both models, the response of the CDS is positive, but not statistically significant at the 95% level.[34] Concerning the other endogenous variables in the extended model, the direction of the effect is similar to the IVLP model in most cases, but the effect is largely smaller and never statistically significant.

Of course, this is consistent with the presence of substantial attenuation bias and, possibly, with bias arising from ignoring non-rating CRA actions. In other terms, the markets seem to largely anticipate CRAs rating decisions as well as react to other CRA actions. Ignoring this seem to lead to substantial underestimation of the impact of CRAs, consistently to what has been found in other contributions (see, e.g., Ismailescu and Kazemi, 2010; Binici and Hutchinson, 2018).



[33] In the case of the baseline model, we only report the responses without global controls. The other versions of the baseline model give similar results that are available upon request.

[34] Quantitatively, the response is slightly larger in the extended model. In principle, this is consistent with expectations being measured better in the extended model. In fact, recall that, in the timing restrictions approach, attenuation bias emerges if rating decisions are anticipated by market participants and market expectations are not measured properly. As explained in section 2a, including global controls and lags of the endogenous variables in regression (23) is a way to measure expectations. In this context, the extended model is more likely to do a good job at measuring expectations, as it includes the lags of more endogenous variables and the global controls. In fact, while the baseline model only includes lags of the CDS, the extended model also includes lags of the CDS of Pemex, the exchange rate, the stock market, the EMBI and the CEMBI spreads as well as the contemporaneous value and lags of the VIX, Brent oil prices and the CDS of other EMEs.

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Figure 6 – impulse response of the sovereign CDS in the baseline timing restriction model, with only the daily change of the CDS as endogenous variable, the rating change as shock and no exogenous control. Ten lags of the daily change of the CDS and nine lags of the rating change are included. The continuous black line represents the impulse response function to a rating downgrade shock. The dashed lines represent the 95% autocorrelation and heteroskedasticity robust Newey-West confidence bands.

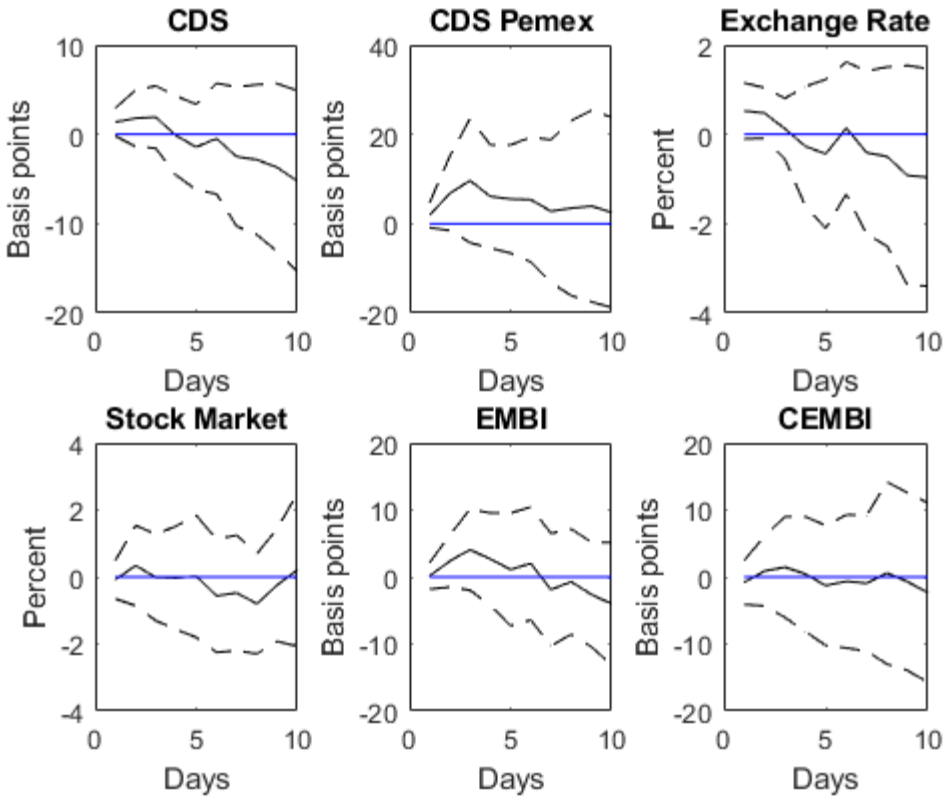

Figure 7 – Impulse responses in the extended timing restriction model with the daily change of the CDS, the CDS of PEMEX, the EMBI+ and CEMBI spreads and the daily change of the log of peso-USD exchange rate and of PCI Mexican stock markets index as endogenous variables and the rating change as shock. The daily change of the average of the CDS of EMEs and the daily change of the log of the VIX and of the Brent oil price are included as controls. Ten lags of the endogenous variables and nine lags of the rating change and of controls are included. The continuous black line represents the impulse response function to a rating downgrade shock. Dashed lines represent the 95% autocorrelation and heteroskedasticity robust Newey-West confidence bands.

d) IV-VAR

Both IV-LPs and IV-VARs can be utilized to merge high-frequency identification with conventional macroeconometric methods. However, they make assumptions that have different consequences in terms of i) statistical power and ii) the trade-off between bias and efficiency.

Regarding i), IV-VARs assume invertibility, meaning that the VAR contains all the relevant information to identify the structural shocks, whereas IV-LPs do not. Under invertibility, it is possible to measure the structural CRA shock, not only on announcement days, but on all days, which results in a higher statistical power. This means that signal-to-noise ratios increase and the effect of CRA shocks can be estimated more accurately, especially for horizons longer than a few days. However, though routinely made, the

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assumption of invertibility is a strong one and can lead to the misattribution of the effect of other shocks to the CRA shock (Stock and Watson, 2018; Känzig, 2021).

Regarding ii), IV-VARs tend to be more efficient but also more biased compared to IV-LPs, with differences that can be significant at intermediate and long horizons (Li et al, 2022). The main reason for these differences is that LPs directly project outcome variables at future horizons onto current covariates, while VARs extrapolate longer-run impulse responses from regressions on the first horizon. Direct projections by LPs are often associated with a large amount of noise, resulting in less efficient estimates. On the other hand, VARs rely on the model being an adequate representation of the dynamics of the variables in the system to extrapolate impulse responses and produce unbiased results at long horizons.

In this section, we evaluate the impact of using an IV-LP by estimating our model using an IV-VAR instead. To mitigate the risks associated with invertibility and bias, it is beneficial to include a larger number of variables in the IV-VAR to address information deficiencies and better describe the data generating process. Therefore, we use the extended model, which contains a larger number of controls and endogenous variables. In practice, we estimate the following reduced form VAR:

$$(18) \qquad \Delta Y_t = \bar{Y} + \sum_{i=1}^{L} C_i \Delta Y_{t-i} + \sum_{i=0}^{L-1} D_i z_{t-i} + u_t$$

where ΔY_t and z_t is the vector of endogenous and exogenous variables respectively; $\bar{Y}$, C_i and D_i are coefficient matrices and u_t is a vector of residuals. We include in ΔY_t and z_t the same endogenous variables and controls of the extended model. As usual in VARs, we postulate a linear relation between the structural shocks and the residuals:

$$(19) \qquad u_t = B\epsilon_t$$

where B is an NxN structural impact matrix and by definition structural shocks are orthogonal, therefore their variance-covariance matrix Ω is diagonal. From (19), it must be the case that

$$(20) \qquad \Upsilon = B\Omega B'$$

where Υ is the variance-covariance matrix of the residuals. Assuming that the structural CRA shock is the first shock in the VAR, our aim is to identify the first column of B, b. In the case of the IV-VAR, the instrument has to satisfy the following conditions:

$i)$ $E\left(\epsilon_{\mathrm{CRA,t}} Z_t^{\perp}\right) = \mathrm{B} \neq 0$ (relevance);

$ii)$ $E\left(\epsilon_{2:n,t} Z_t^{\perp}\right) = 0;$ (exogeneity).

Similar to the IV-LP model, the instrument must be correlated with the structural CRA shock (the relevance condition) and uncorrelated with other structural shocks (the exogeneity condition). In contrast to the IV-LP, in the IV-VAR, the lead-lag exogeneity condition is redundant, as it is implied by invertibility.

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Under assumptions i) and ii), the contemporaneous response of the endogenous variables relative to the CDS can be obtained as follows:

$$(21) \qquad \frac{b_{2:n}}{b_1} = \frac{E(u_{2:n,t}Z_t^{\perp})}{E(u_{1,t}Z_t^{\perp})}.$$

As in the IV-LP case, the scale of b_1 is normalized to one basis point. Contrary to the IV-LP, (20) can be used together with (21) to obtain the structural CRA shock on all days. To ensure that condition ii) is met, we orthogonalize the instrument regressing it against ten lags of the endogenous variables and the instrument and the contemporaneous value and nine lags of the exogenous global controls.

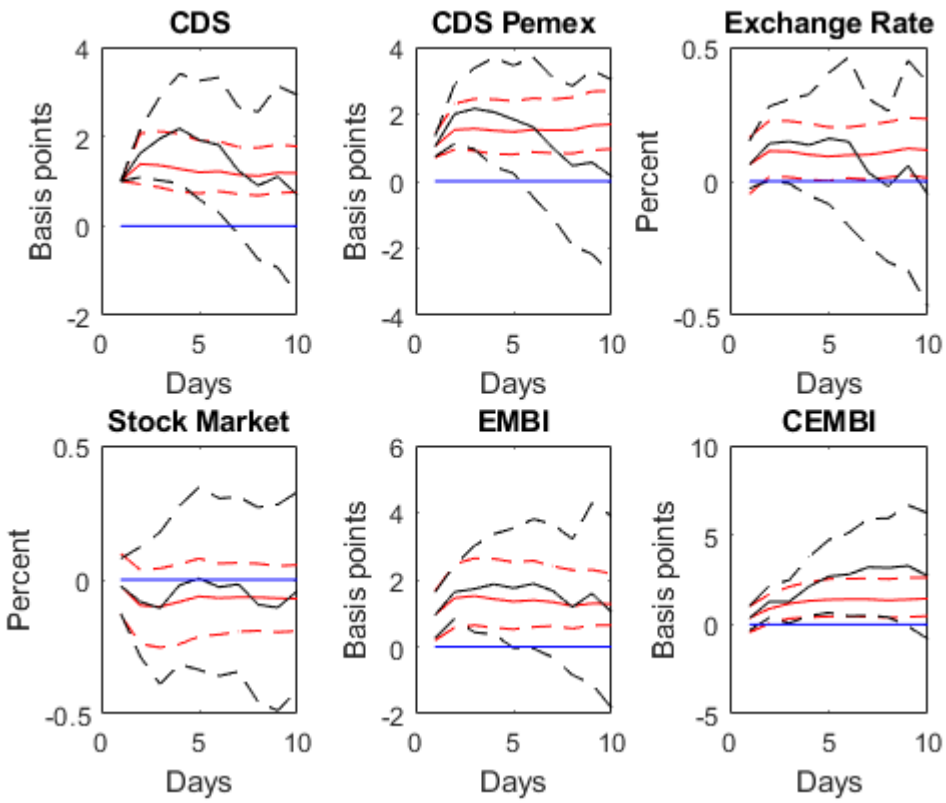

Figure 8 – Impulse responses in the extended IV-VAR (red lines) and IV-LP (black lines) models with the daily change of the CDS, the CDS of PEMEX, the EMBI+ and CEMBI spreads and the daily change of the log of peso-USD exchange rate and of PCI Mexican stock markets index as endogenous variables. The daily change of the average of the CDS of EMEs and the daily change of the log of the VIX and of the Brent oil price are included as controls. Ten lags of the endogenous variables and nine lags of controls are included. The continuous red (black) line represents the impulse response function after a CRA shock that generates an increase in the Sovereign CDS of 1 basis point in the IV-VAR (IV-LP) model. Dashed red lines represent the 95% confidence bands of the IV-VAR computed with the block-bootstrap method proposed by Jentsch and Lunsford (2022). Dashed black lines represent the 95% autocorrelation and heteroskedasticity robust Newey-West confidence bands of the IV-LP.

Figure 8 shows the impulse responses in the IV-VAR (in red) and in the IV-LP (in black). There are some quantitative differences: the response of the sovereign CDS, the CDS of Pemex and the exchange rate is smaller in the first few days and larger later in the IV-VAR;

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while that of the CEMBI spread is smaller at all horizons. The responses of the stock market and the EMBI spread is similar in the two models.

However, the most striking differences that emerge concern inference: confidence sets are much larger in the IV-LP than in the IV-VAR, especially at longer horizons, consistently with the IV-VAR being more efficient. For this reason, in the IV-VAR the impulse responses are statistically significant for longer: in particular, the response of all variables, apart from the stock market, is still significant after two weeks.

e) Results including only dates with rating or outlook changes

As explained above, one of the main advantages of the high-frequency IV-LP approach is to capture the effect of non-rating CRA shocks. This, in turn, implies that, in contrast to the timing restriction approach, we can use also days in which CRAs confirm the rating to measure their effect on the markets. Still, the CRA signal on these days may be smaller than on days in which the rating or the outlook are changed, meaning that the risk of the CDS being driven by other shocks on these days may be larger.

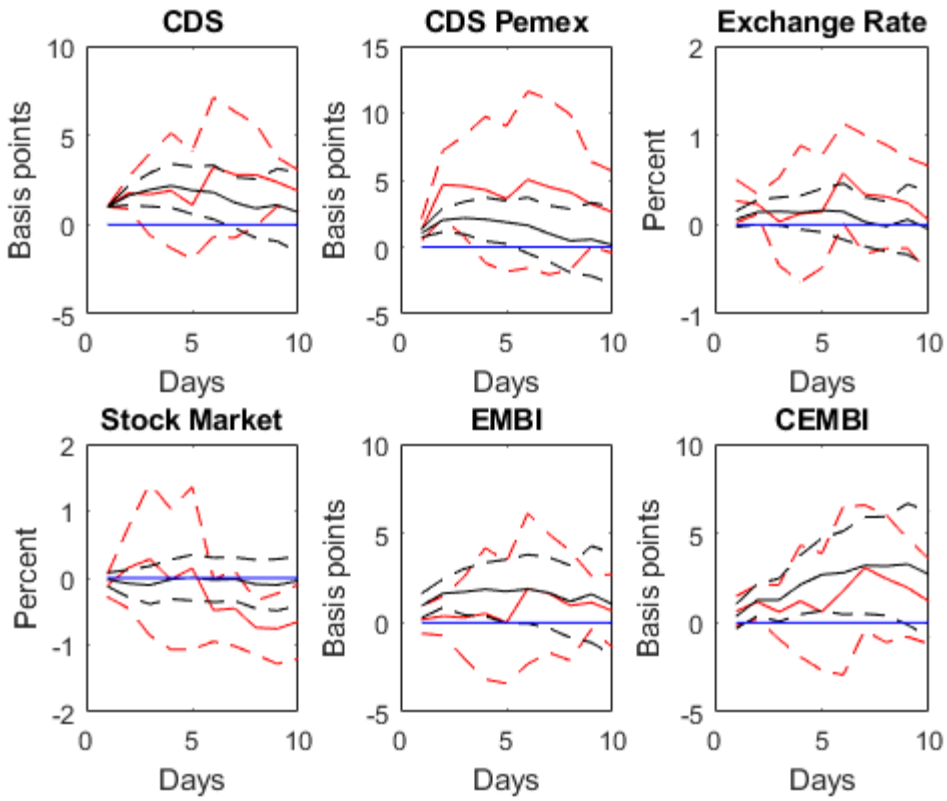

Figure 9 – Impulse responses in the extended IV-LP including only days with rating or outlook changes (red lines) and including all CRA announcements (black lines) in the instrument with the daily change of the CDS, the CDS of PEMEX, the EMBI+ and CEMBI spreads and the daily change of the log of peso-USD exchange rate and of PCI Mexican stock markets index as endogenous variables. The daily change of the average of the CDS of EMEs and the daily change of the log of the VIX and of the Brent oil price are included as controls. Ten lags of the endogenous variables and nine lags of controls are included. The continuous red line represents the impulse response function after a CRA shock that generates an increase in the Sovereign CDS of 1 basis point in the model with only days with rating or outlook changes in the instrument. The continuous black line represents the impulse response function after a CRA shock that generates an increase in the

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Sovereign CDS of 1 basis point in the model with all announcement days in the instrument. Dashed lines represent the 95% autocorrelation and heteroskedasticity robust Newey-West confidence bands.

Therefore, in this section, we re-estimate the IV-LP model, including in the instrument only the days in which either a rating or an outlook change is made. Results are reported in Figure 9,[35] which shows the impulse responses (in red) and compares them to the impulse responses when all announcement days are included (in black).[36] The number of days included in the instrument is smaller, which reduces statistical power, making confidence bands larger and impulse responses noisier. However, results are similar, with both the direction of the response and the statistical significance being the same for most variables. The only exceptions are the EMBI spread and the stock market.

In particular, in contrast to the main specification where the EMBI increases, its response is not statistically significant. The opposite is true in the case of the stock market: while in the main specification its response is not statistically significant, it is when only rating or outlook change days are included in the instrument. More specifically, the stock market falls after a few days, with the reduction being statistically significant in the second week after the shock.

6. Conclusion

The ability of CRAs to influence financial markets is a widely debated topic. Some commentators argue that CRAs are able to affect market assessment of sovereign and corporate risk, as well as have an impact on the stock and FX markets. Others, on the other hand, reckon that CRAs mainly follow investor opinion and, therefore, have limited effects on the markets (Brealey and Myers, 2003).

To understand if CRAs' actions have relevant effects on financial markets, in this paper, we have employed the IV-LP approach, which has already been used to achieve credible identification of monetary policy, fiscal policy, oil supply and political risk shocks (Gertler and Karadi, 2015; Mertens and Ravn, 2013; Känzig, 2021; Balduzzi et al, 2023). This approach allows us to identify the effect of structural shocks to CRAs' communication about the Mexican sovereign on financial markets. In particular, we use as instrument the change of the sovereign CDS on days of CRAs' rating announcements. The small window around the event and the inclusion of several domestic and global financial variables as controls, in fact, this variable only responds to CRAs' announcements, satisfying the relevance, contemporaneous and lead-lag exogeneity requirements for an instrument. With this technique, we show that CRAs' actions have statistically significant effects not only on sovereign CDS spreads and interest rates, but also on the funding cost of the private sector, as well as on the exchange rate.

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[35] We present only the results of the extended model, but those of the baseline model are similar.
[36] The robust F statistics is 19.67.

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Appendix

List of Emerging and Middle-Income Countries

The average CDS spread of emerging markets and middle-income countries is calculated following the definition of the International Monetary Fund presented in its *Fiscal Monitor*. From this list, we use the 22 countries for which there is available their five-year contract CDS Spread in US dollars on IHS Markit. We use only the countries that have data available from the starting date of our sample, January 4, 2010, in order to avoid distortions in the average value from incomplete data series. These countries and their tickers are presented in the next table.

Country	IHS Markit Ticker
Poland	POLAND\SNRFOR\USD\CR14\5Y\100

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Romania	ROMANI\SNRFOR\USD\CR14\5Y\100
Croatia	CROATI\SNRFOR\USD\CR14\5Y\100
Kazakhstan	KAZAKS\SNRFOR\USD\CR14\5Y\100
Russian Federation	RUSSIA\SNRFOR\USD\CR14\5Y\100
Brazil	BRAZIL\SNRFOR\USD\CR14\5Y\100
Chile	CHILE\SNRFOR\USD\CR14\5Y\100
Colombia	COLOM\SNRFOR\USD\CR14\5Y\100
Dominican Republic	DOMREP\SNRFOR\USD\CR14\5Y\100
Turkey	TURKEY\SNRFOR\USD\CR14\5Y\100
South Africa	SOAF\SNRFOR\USD\CR14\5Y\100
Saudi Arabia	SAUDI\SNRFOR\USD\CR14\5Y\100
Peru	PERU\SNRFOR\USD\CR14\5Y\100
Indonesia	INDON\SNRFOR\USD\CR14\5Y\100
Philippines	PHILIP\SNRFOR\USD\CR14\5Y\100
Thailand	THAI\SNRFOR\USD\CR14\5Y\100
Uruguay	URUGAY\SNRFOR\USD\CR14\5Y\100
Egypt	EGYPT\SNRFOR\USD\CR14\5Y\100
Qatar	QATAR\SNRFOR\USD\CR14\5Y\100
United Arab Emirates	UAE\SNRFOR\USD\CR14\5Y\100
China	CHINA\SNRFOR\USD\CR14\5Y\100
Malaysia	MALAYS\SNRFOR\USD\CR14\5Y\100

Placebo test

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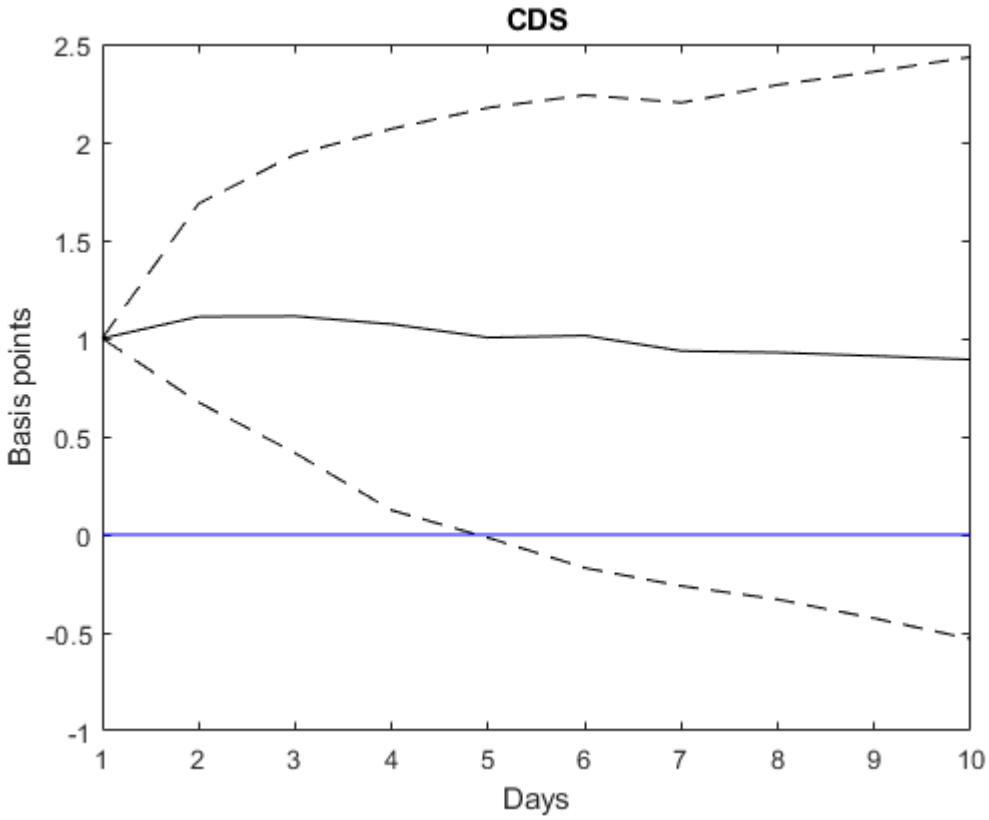

Figure A1 – Placebo test. Impulse response of the sovereign CDS in the baseline model, with only the daily change of the CDS as endogenous variable and no exogenous control. Ten lags of the daily change of the CDS and nine lags of the instrument are included. The model is estimated using as instrument the daily change in the sovereign CDS spread on randomly selected dates. We repeat this procedure 2000 times. The dashed lines are the 97.5 and 2.5 percentiles.

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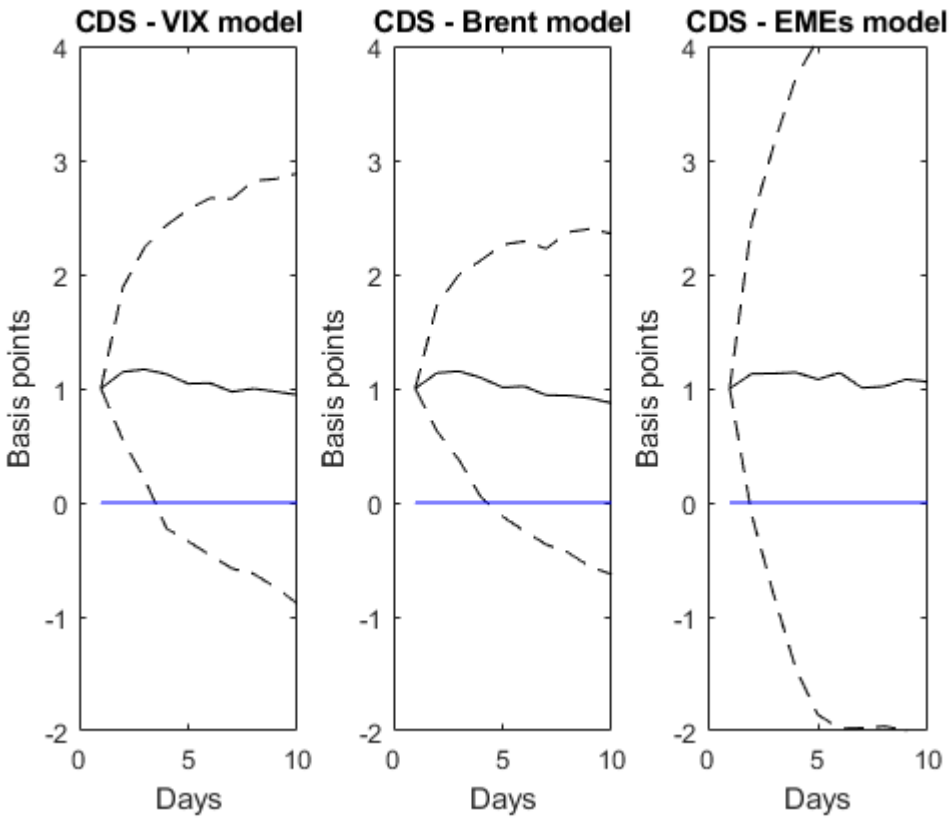

Figure A2 – Placebo test. The left panel reports the impulse response function of the sovereign CDS in the model with the contemporaneous and nine lags of the daily change in the log of the VIX as controls and only the daily change of the CDS as endogenous variable. The central panel reports the impulse response function of the sovereign CDS in the model with the contemporaneous and nine lags of the daily change in the log of Brent price as controls and only the daily change of the CDS as endogenous variable. The right panel reports the impulse response function of the sovereign CDS in the model with the contemporaneous and nine lags of the daily change of an average of the CDS of EMEs as controls and only the daily change of the CDS as endogenous variable. In all models, ten lags of the daily change of the CDS and nine lags of the instrument are included. The model is estimated using as instrument the daily change in the sovereign CDS spread on randomly selected dates. We repeat this procedure 2000 times. The dashed lines are the 97.5 and 2.5 percentiles.

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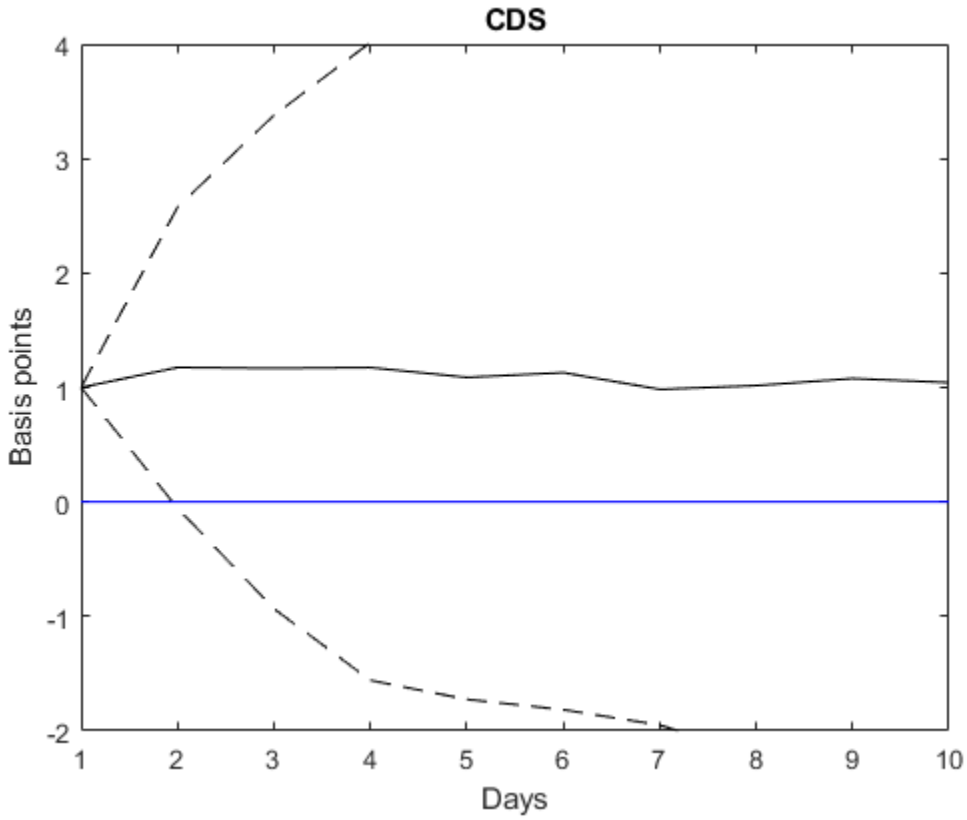

Figure A3 – Placebo test. Impulse response of the sovereign CDS in the model with the daily change of the log of the VIX, the daily change of the log of Brent oil price and the daily change of the average CDS of EMEs as controls and only the daily change of the CDS as endogenous variable. Nine lags of the controls and of the instrument and ten lags of the CDS are included. The model is estimated using as instrument the daily change in the sovereign CDS spread on randomly selected dates. We repeat this procedure 2000 times. The dashed lines are the 97.5 and 2.5 percentiles.

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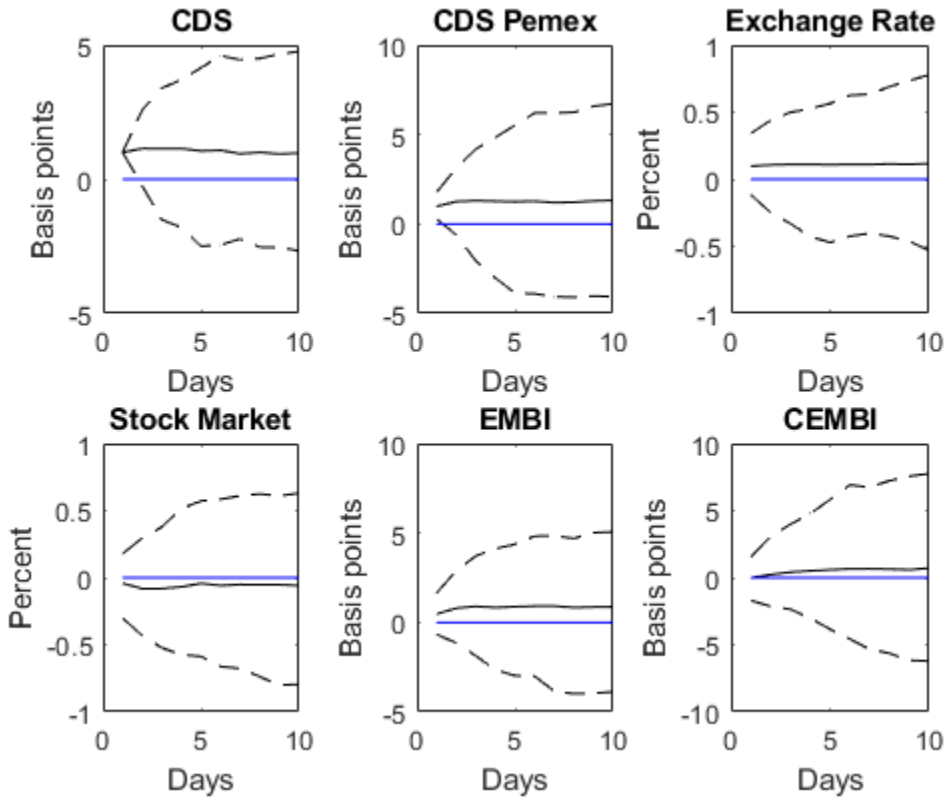

Figure A4 – Placebo test. Impulse responses in the extended model with the daily change of the CDS, the CDS of PEMEX, the EMBI+ and CEMBI spreads and the daily change of the log of peso-USD exchange rate and of PCI Mexican stock markets index as endogenous variables. The daily change of the average of the CDS of EMEs and the daily change of the log of the VIX and of the Brent oil price are included as controls. Ten lags of the endogenous variables and nine lags of the controls and of the instrument are included. The continuous black line represents the impulse response function after a CRA shock that generates an increase in the Sovereign CDS of 1 basis point. The model is estimated using as instrument the daily change in the sovereign CDS spread on randomly selected dates. We repeat this procedure 2000 times. The dashed lines are the 97.5 and 2.5 percentiles.

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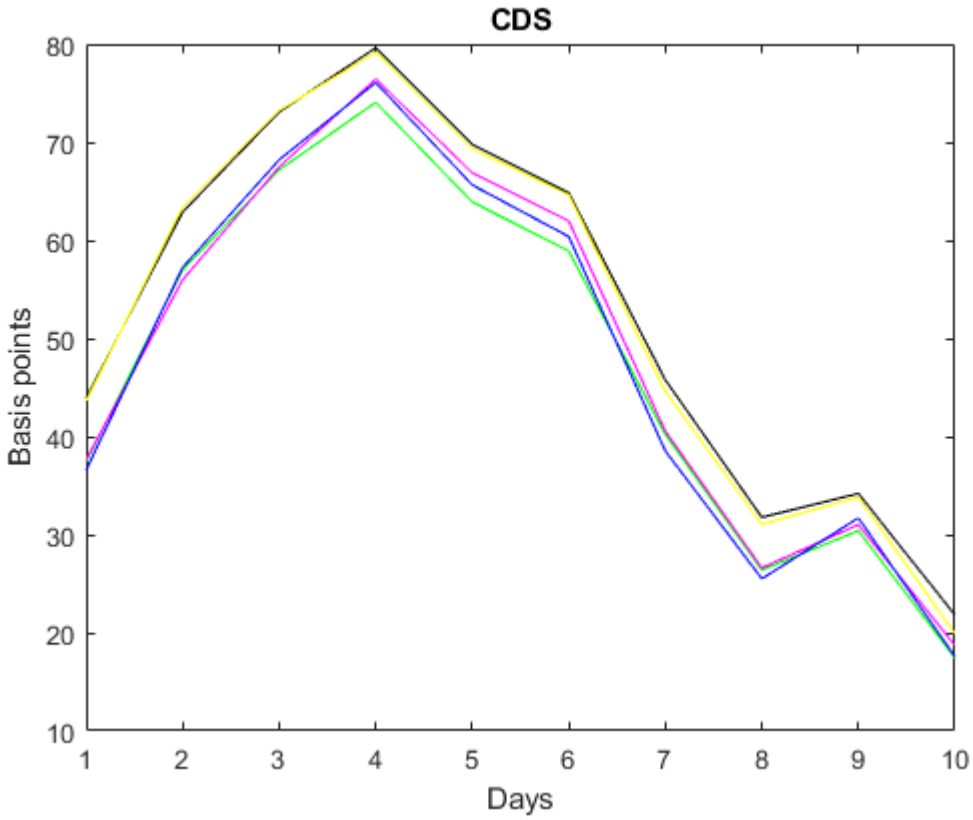

Figure A5 – impulse response of the sovereign CDS in the baseline model with the CDS and the rating as endogenous variables, without exogenous controls (black line), the VIX as control (green line), the Brent as control (yellow line), the CDS of EMEs as control (purple line) and the three exogenous controls at the same time (blue line). The shock is an increase of 1 in the rating variable which corresponds to a downgrade. Ten lags of the daily change of the CDS and the rating and nine lags of the instrument are included.

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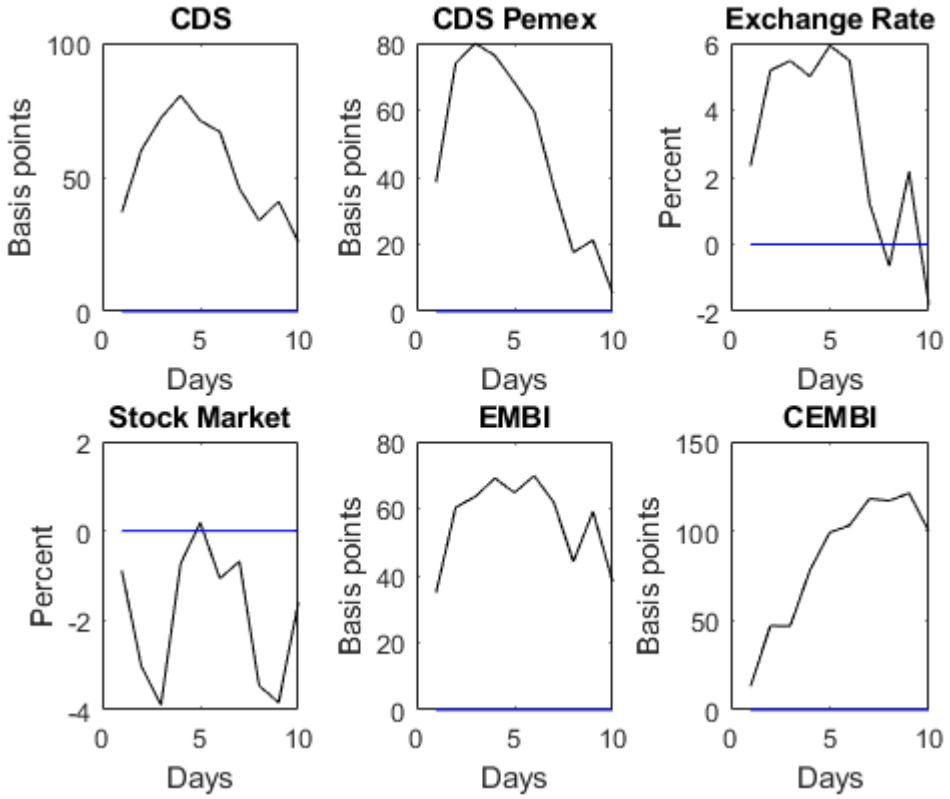

Figure A6 – Impulse responses in the extended model with the daily change of the CDS, the rating, the CDS of PEMEX, the EMBI+ and CEMBI spreads and the daily change of the log of peso-USD exchange rate and of PCI Mexican stock markets index as endogenous variables. The daily change of the average of the CDS of EMEs and the daily change of the log of the VIX and of the Brent oil price are included as controls. Ten lags of the endogenous variables and nine lags of the controls and of the instrument are included. The shock is an increase of 1 in the rating variable which corresponds to a downgrade.

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